BANK CUSTOMER CHURN ANALYSIS

End-to-End Data Analytics Project

Tools: Python • SQL / PostgreSQL • Power BI

Focus: Churn patterns, customer segments, recorded balance exposure and retention opportunities

Business question

Which customer segments are associated with higher churn, how much recorded balance belongs to churned customers, and where should the bank investigate retention opportunities?

Prepared as a Data Analyst portfolio project. This report uses a static customer snapshot and does not claim to predict future churn or prove causal drivers.

1. Executive Summary

This project analyzes 10,000 bank-customer records using Python for exploratory data analysis, SQL for repeatable business analysis, and Power BI for interactive reporting. The supplied results show 2,037 churned customers and an overall churn rate of 20.37%. The recorded balances of churned customers total approximately $185.59 million, about 24.26% of total customer balance.

Metric	Result	Meaning
Total customers	10,000	Customer records analyzed
Churned customers	2,037	Records with churn = 1
Overall churn rate	20.37%	Share of customers marked churned
Average credit score	650.53	Average across customers
Average balance	$76,485.89	Mean recorded customer balance
Average estimated salary	$100,090.24	Mean estimated salary
Churned customer balance	$185.59M	Balance sum for customers marked churned

Main conclusion: compare churn rate, segment size and churned balance together. A high churn percentage in a very small group should not automatically become the bank's top retention priority.

2. Business Problem & Objectives

Business problem: The bank wants to understand which customer segments have higher observed churn, what customer characteristics are associated with churn, and which segments warrant further retention investigation.

Objectives

• Measure overall churn and the number of churned customers.
• Compare churn by age, gender, country, tenure, product count, activity and financial bands.
• Quantify recorded balance associated with churned customers.
• Identify segments that combine higher churn with meaningful customer volume or balance.
• Present the analysis in a Power BI report for business stakeholders.

Dataset fields: customer_id, credit_score, country, gender, age, tenure, balance, products_number, credit_card, active_member, estimated_salary and churn. The dataset contains 10,000 unique customer IDs and 12 columns, according to the supplied notebook review.

Scope limitation: The data does not include churn dates, transaction timelines, monthly snapshots, complaints, customer-service interactions or campaign results. Therefore, this project is descriptive, not a 30-day prediction or causal analysis.

3. Tools & Workflow

Python: data inspection, missing-value and uniqueness checks, feature grouping and exploratory comparisons of churn by customer attributes.
SQL / PostgreSQL: aggregations, KPI calculations, country comparisons, product/activity analysis, financial bands and multi-dimensional segmentation.
Power BI: KPI cards, segment visuals, filters and a two-page report to communicate churn patterns and retention opportunities.

Workflow: Kaggle CSV → Python EDA → SQL business analysis → Power BI report → findings and recommendations.

4. Python EDA — Key Findings

Python was used to inspect the dataset, check missing values and customer ID uniqueness, examine distributions, create age groups, compare churn across dimensions, and calculate the balance associated with churned customers. The goal was to discover patterns without overcomplicating the EDA.

Dimension	Observed result	Interpretation
Age 46–55	50.57% churn	Highest observed churn rate among the age groups shared
Age 56–65	48.32% churn	Also elevated relative to younger age groups
Activity	Inactive 26.85%; active 14.27%	12.58 percentage-point difference
Country	Germany 32.44%; Spain 16.67%; France 16.15%	Germany warrants further investigation
Gender	Female 25.07%; male 16.46%	Observed difference; not evidence of causation
Credit card	No 20.81%; yes 20.18%	Small standalone difference
Balance exposure	$185.59M churned-customer balance	Historical balance association, not lost profit

Product-count caveat: churn rates were 27.71% for one product, 7.58% for two, 82.71% for three and 100% for four. The three- and four-product groups have only 266 and 60 customers, respectively, so those high rates should be interpreted alongside their small sample sizes.

5. SQL Analysis

SQL translated the business problem into repeatable queries. The supplied scripts cover basic KPIs, demographics, geography, product holdings, engagement, financial attributes and multi-dimensional segments.

Query groups completed

1. Basic KPIs: total customers, churned customers, churn rate, average credit score, balance and salary, active-member percentage.
2. Demographics: churn by gender and age group; age group with most churned customers.
3. Geography: country churn rates and customer-base share.
4. Products: churn by product count, credit-card ownership and product/activity combinations.
5. Engagement: churn among active/inactive customers, inactive customers by country and age.
6. Financial: churn by credit-score, balance and salary bands; churned balance.
7. Segmentation: country, age group, activity and product-count combinations.

Key SQL results

Metric	Result
Total customers	10,000
Churned customers / churn rate	2,037 / 20.37%
Germany churn rate	32.44%
Female / male churn rate	25.07% / 16.46%
Salary bands	Roughly 19.87%–21.47% in supplied results
Balance bands	Highest shown: 200K+ at 55.88%; other bands vary

SQL quality note: use COUNT(*) FILTER (WHERE churn = 1) or SUM(churn) to count churned customers; COUNT(churn) counts all non-null churn values. Segment churn rate should be calculated as churned customers divided by total customers in the same segment.

6. Business Interpretation

1. Churn is material. One in five customers in the snapshot is marked churned, making churn a clear KPI for ongoing monitoring.

2. Historical balance exposure is significant. Churned customers account for about 24.26% of total recorded balance. This is not equivalent to lost revenue, profit or a forecast of future losses.

3. Inactive customers warrant investigation. Their observed churn rate is higher than that of active customers. The static dataset cannot establish whether inactivity occurred before churn.

4. Older age groups show higher churn. The 46–55 and 56–65 segments should be compared by activity, country, product count and balance to see whether the pattern persists.

5. Germany warrants further investigation. Compare similar customer profiles across countries before attributing the difference to geography itself.

6. Product-count patterns are non-linear. Two-product customers show lower churn than one-product customers, while three- and four-product groups have high rates but small populations.

7. Salary band alone appears less discriminative. The supplied salary-band churn rates are relatively close, so salary should not dominate retention prioritization based on this analysis alone.

7. Power BI Report

The two-page report separates the broad churn overview from retention opportunities. The overview uses KPI cards and charts to communicate churn by age, country, products and activity. The retention page ranks customer segments and displays customers, churned customers, churn rate, churned balance and priority labels.

Dashboard screenshots are included in the following pages.

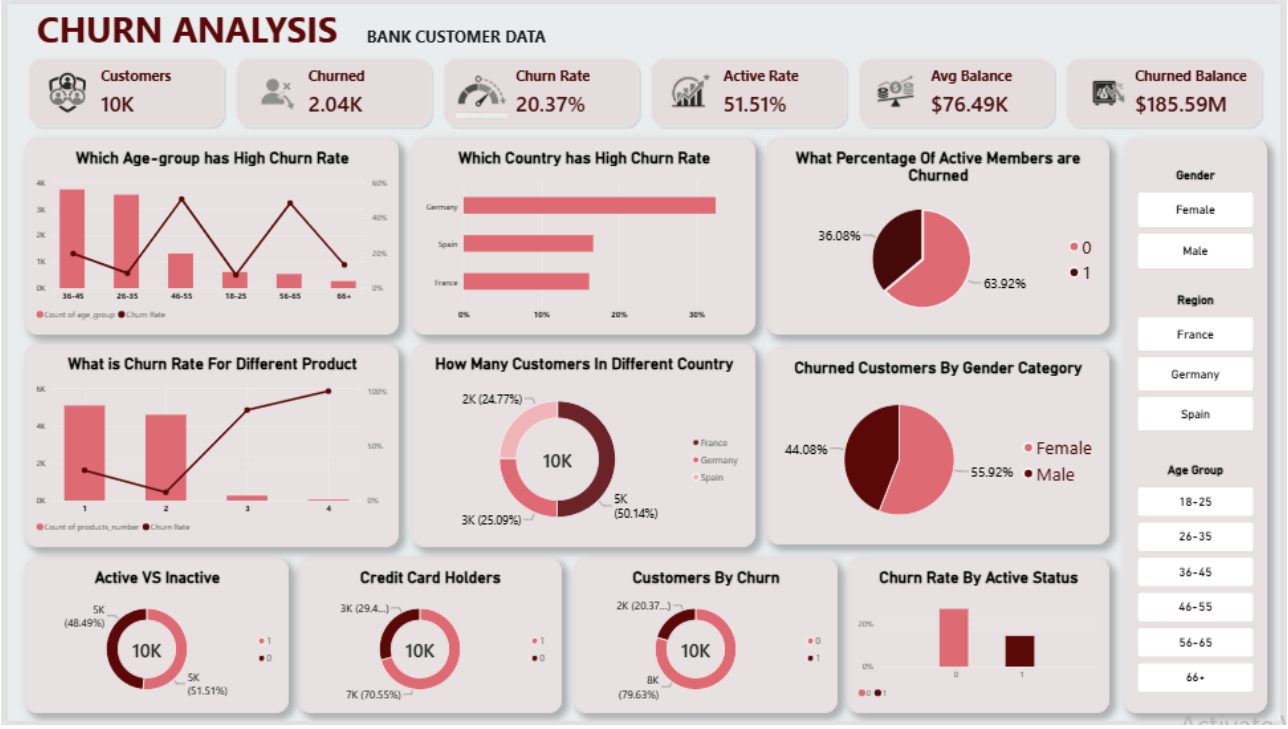


Figure 1. Final Power BI report — Customer Churn Overview

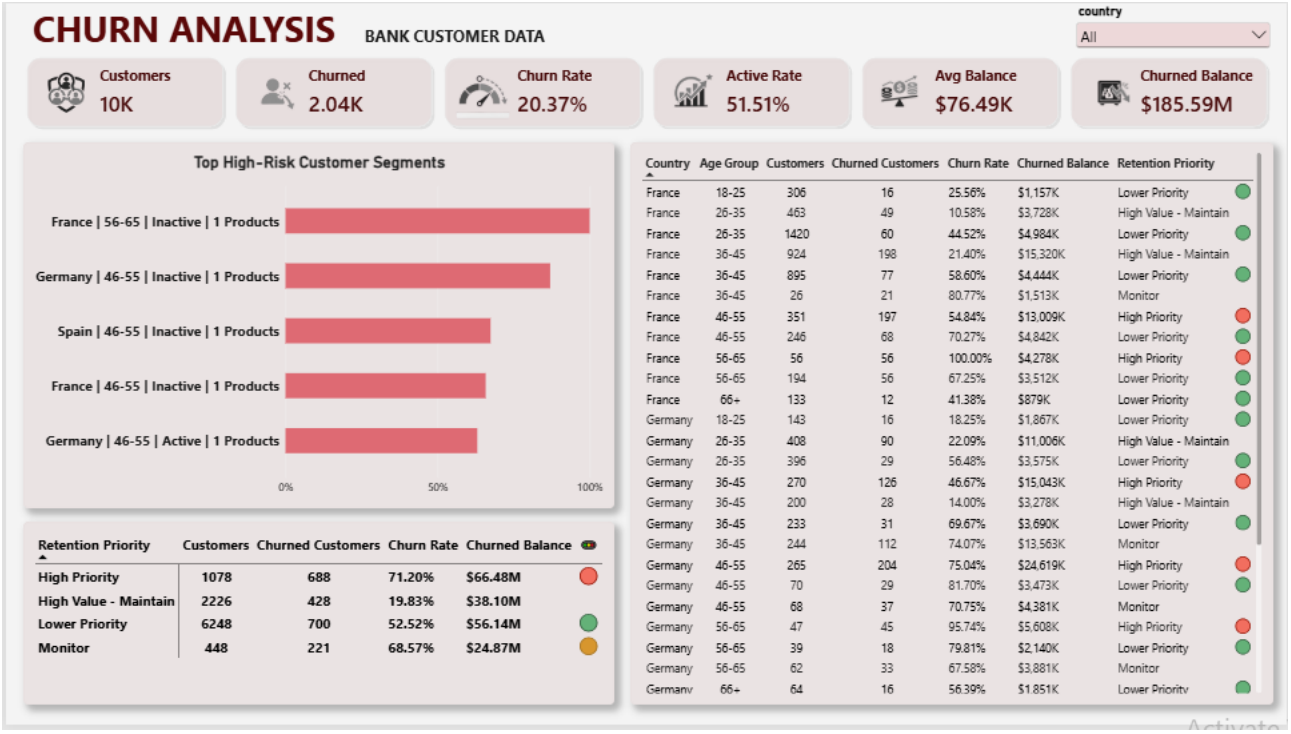
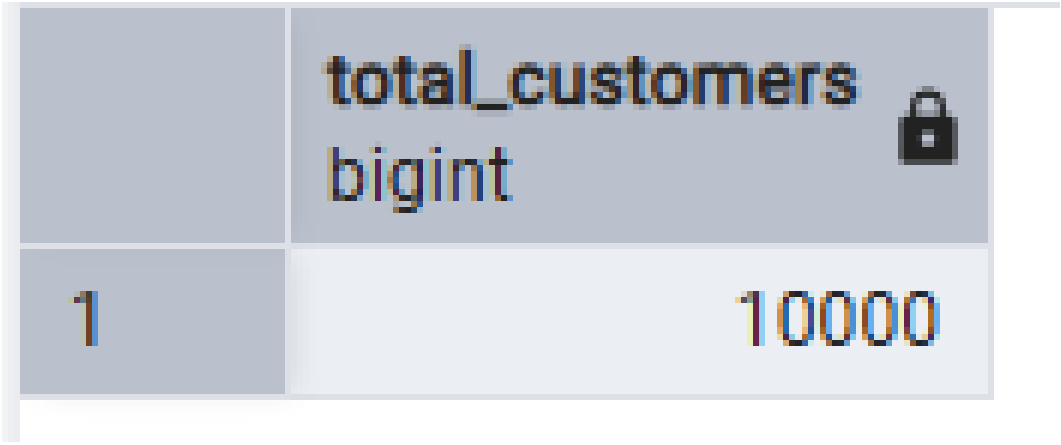


Figure 2. Final Power BI report — Retention Opportunities

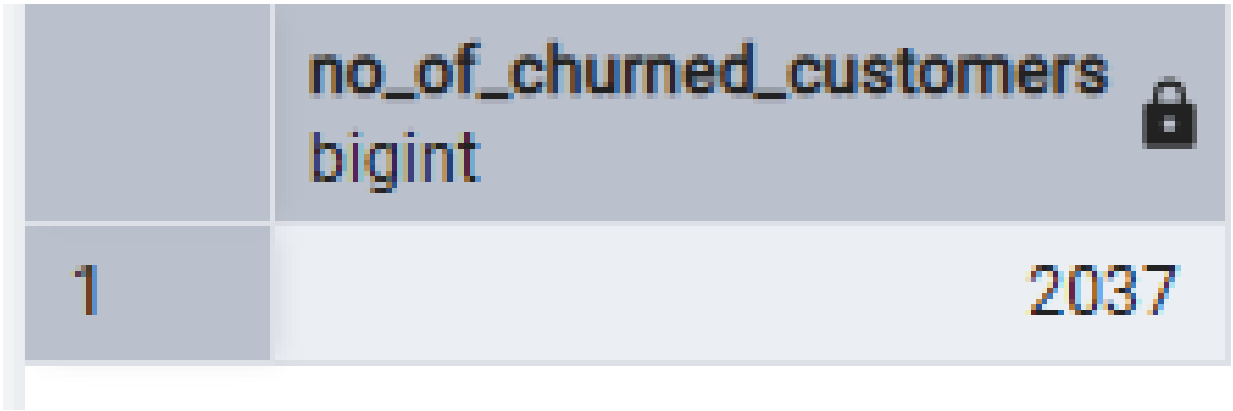
8. SQL Screenshot Evidence

Screenshots supplied for SQL query results are included below as evidence of the analysis performed. Each image caption describes the result represented.



	total_customers bigint
1	10000

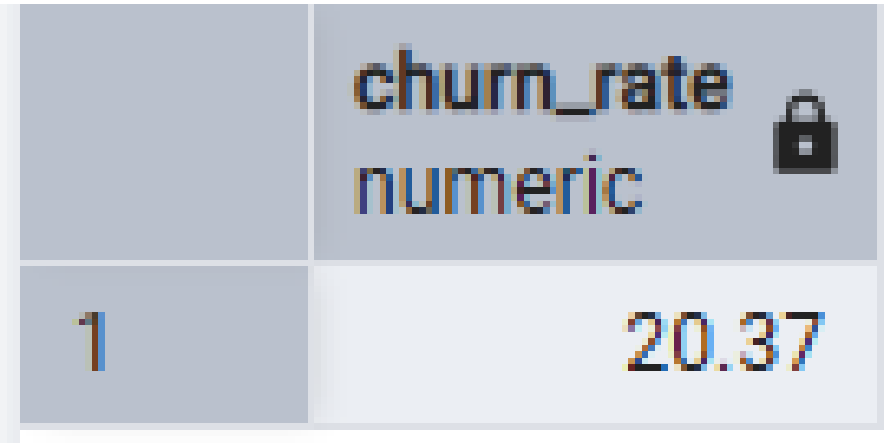
SQL: total customers



	no_of_churned_customers bigint
1	2037

SQL: churned customers

SQL Screenshot Evidence (continued)



	churn_rate numeric
1	20.37

SQL: overall churn rate

	avg_credit_score numeric	avg_balance numeric	avg_salary numeric
1	650.53	76485.89	100090.24

SQL: average credit score, balance and salary

SQL Screenshot Evidence (continued)

	churn_rate numeric	gender text
1	25.07	Female
2	16.46	Male

SQL: churn rate by gender

	churn_rate numeric	age_group text
2	19.62	36-45
3	50.57	46-55
4	48.32	56-65
5	7.47	18-25
6	13.26	66+
7	8.50	26-35

SQL: churn rate by age group

SQL Screenshot Evidence (continued)

	country text	total_customers bigint	percentage_of_total_base numeric	country_churn_rate numeric
1	France	5014	50.14	16.15
2	Germany	2509	25.09	32.44
3	Spain	2477	24.77	16.67

SQL: customer distribution and churn by country

	churn_rate numeric	products_number bigint
1	27.71	1
2	82.71	3
3	100.00	4
4	7.58	2

SQL: churn rate by product count

SQL Screenshot Evidence (continued)

	product_segment bigint	total_customers bigint	churn_rate numeric
1	4	60	100.00
2	3	266	82.71
3	1	5084	27.71
4	2	4590	7.58

SQL: customer count by product count

	customer_no bigint	active_member bigint	products_number bigint
1	2563	1	1
2	2521	0	1
3	2446	1	2
4	2144	0	2
5	113	1	3
6	153	0	3
7	29	1	4
8	31	0	4

SQL: products by activity status

SQL Screenshot Evidence (continued)

	churn_rate numeric	active_member bigint	age_group text
1	93.85	0	56-65
2	63.61	0	46-55
3	62.50	0	66+
4	24.22	0	36-45
5	11.47	0	18-25
6	10.80	0	26-35

SQL: inactive customers by age group

	churn_rate numeric	active_member bigint	products_number bigint
1	18.92	1	1
2	36.65	0	1
3	5.56	1	2
4	9.89	0	2
5	88.24	0	3
6	75.22	1	3
7	100.00	0	4
8	100.00	1	4

SQL: churn by activity and product count

SQL Screenshot Evidence (continued)

	churn_rate numeric	balance_band character varying (20)
1	55.88	200k+
2	34.67	1–50k
3	19.88	50k–100k
4	21.93	150k–200k
5	13.82	0
6	25.77	100k–150k

SQL: churn by balance band

	churn_rate numeric	salary_band character varying (20)
1	19.93	<50k
2	19.87	50k–100k
3	21.47	150k–200k
4	20.23	100k–150k

SQL: churn by salary band

SQL Screenshot Evidence (continued)

	country text	customer bigint	products_number bigint	active_member bigint	age_group text	churn_rate numeric
1	France	5	3	1	56-65	100.00
2	Germany	3	4	0	46-55	100.00
3	Germany	4	3	1	56-65	100.00
4	Germany	1	4	1	26-35	100.00
5	Spain	1	4	1	26-35	100.00
6	Germany	2	3	0	66+	100.00
7	France	56	1	0	56-65	100.00
8	Germany	3	4	0	56-65	100.00
9	Germany	6	3	0	56-65	100.00
10	France	16	3	0	46-55	100.00

SQL: churn across customer segments

9. Business Recommendations

1. Investigate inactive customers. Compare their churn rate across country, age, product count and balance. With time-based activity data, test whether falling engagement precedes churn.

2. Investigate customers aged 46–65. Review product relationships, service needs and balance distribution instead of assuming age itself causes churn.

3. Investigate the Germany gap. Compare similar German, French and Spanish customer segments to determine whether the gap remains after accounting for activity, age and product count.

4. Review product relationships. Examine one-product customers and investigate three-/four-product segments with small sample sizes before recommending product changes or cross-selling.

5. Prioritize segments using multiple measures. Consider churn rate, customer count and churned balance together. Document the thresholds and minimum segment-size rule behind priority labels.

6. Validate retention initiatives. Run measured retention campaigns and compare outcomes against an appropriate comparison group before claiming an intervention reduces churn.

10. Limitations & Next Steps

Limitations: The dataset is a static snapshot, with no churn dates, longitudinal activity, transaction history, complaints or campaign results. It cannot support 30-day pre-churn signals, individual future churn predictions, causal conclusions or direct lost-profit calculations.

Next steps:
- Reconcile Power BI customer counts, churned counts, churn rates and churned balances against SQL.
- Treat priority categories as transparent business heuristics, not predicted risk scores.
- Add a GitHub README, data dictionary, Python notebook and SQL scripts.
- If available, add monthly customer snapshots and activity history for true 30/60/90-day churn monitoring.

11. Final Conclusion

This project demonstrates a complete descriptive analytics workflow: Python for discovery, SQL for structured business analysis, and Power BI for communicating findings. The results show material churn and notable differences across activity, age, country and product-count groups. The recommended business response is to investigate high-churn segments while considering customer volume and recorded balance, then validate retention actions using time-based data and measured outcomes.

Prepared using user-provided SQL result screenshots, Power BI screenshots, and prior notebook review. Interpretations are limited to the supplied evidence.